



Universität
Zürich^{UZH}



Blockchain & DLT
Research Group



Network Science HS24

Assignment 6

Blockchain & Distributed Ledger Technologies Group

FOR STUDIES PURPOSES ONLY

UZH Blockchain Center
Faculty of Business, Economics and Informatics
University of Zurich
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1 Preferential Attachment

Task: Build networks with the Barabási-Albert model. Connect $m = 3$ for each new node and grow until $N = 50, 100, 500, 1000, 5000$, then compute some network properties. Compare them with randomized versions of the networks.

1. (1 point) Compute the average clustering coefficient, assortativity, average shortest path length, and diameter.
2. (1 point) Compare them by scatterplots with the same measures on randomized versions of the networks (use log x-axis).
3. (1 point) Which measures are relatively unchanged by randomization? Why?

2 Scale-free Distribution

For each network dataset provided:

1. (1 point) Compute the exponent γ of the power-law distribution fit $p(k) \sim k^{-\gamma}$ of the network degree distribution and the corresponding error.

Hint: In Python you can fit a power law distribution using the `power law` package.

2. (1 point) Produce a single plot showing:

- the empirical degree distribution,
- the power-law fit,
- the Poisson fit,
- the exponential distribution with mean value equal to $\langle k \rangle$.

Which distribution is more likely to describe the data?

Hint: Make sure the data is presented correctly, use a logarithmic scale when needed.

Hint: The Poisson distribution with mean value $\langle k \rangle = \lambda$ can be computed as:

$$p(k, \lambda) = \frac{\lambda^k}{k!} e^{-\lambda}$$

Hint: The exponential distribution with mean value $\langle k \rangle = \lambda$ can be computed as:

$$p(k, \lambda) = \frac{1}{\lambda} e^{-\frac{k}{\lambda}}$$

Hint: For the factorial $k!$ you can use the factorial function that can be imported from the module `scipy.special`

3. (1 point) Based on the previous plot, say whether the network under consideration is scale-free or not.

Datasets provided

- `graph_macaque.gml`: Macaque dominance Network. This network contains dominance behavior in a colony of 62 adult female Japanese macaques Takahata [1991]. A node represents a macaque, and an edge (A, B) represents a dominance event from A to B. For simplicity, it is provided the undirected version of the graph.

- `graph_internet.gml`: Internet AS graph. Nodes represent autonomous systems, and Edges represent connections between them, Karrer et al. [2014].
- `graph_actors.gml`: Nodes represent actors and Edges represent appearances in the same movie Barabási and Albert [1999].
- `graph_flavors.gml`: Nodes are ingredients and links are created when two ingredients share a common flavor compound Ahn et al. [2011].

References

- Ahn, Y.-Y., Ahnert, S. E., Bagrow, J. P. and Barabási, A.-L. [2011]. Flavor network and the principles of food pairing, *Scientific reports* **1**(1): 196.
- Barabási, A.-L. and Albert, R. [1999]. Emergence of scaling in random networks, *science* **286**(5439): 509–512.
- Karrer, B., Newman, M. and Zdeborova, L. [2014]. Percolation on sparse networks, *Phys. Rev. Lett.* **113**: 208702.
- Takahata, Y. [1991]. Diachronic changes in the dominance relations of adult female Japanese monkeys of the Arashiyama B group, in L. M. Fedigan and P. J. Asquith (eds), *The Monkeys of Arashiyama: Thirty-five Years of Res. in Japan and the West*, pp. 123–139.